

Additional Services Costing: Varsity Cyber Clash (UTeM FTMK)

1. Project Context and Objective

This procurement report defines the strategic architectural requirements for supplemental network services supporting the "Varsity Cyber Clash" regional e-sports tournament at the Fakulti Teknologi Maklumat dan Komunikasi (FTMK), UTeM. As Senior IT Infrastructure Architects, our core objective is to execute a **Value-Engineering** exercise: integrating advanced services that maximize the "Complexity" and "User Experience" scores of the project while adhering to the institutional directive of "minimal cost." By deploying these services, we ensure a robust, high-performance environment that maintains competitive integrity through the proactive mitigation of transmission impairments.

2. Software Licenses for Tournament Operations

Service Category	Software Specification	Cost Justification
Tournament Management System	Specialized Registration & Bracket Logic	Streamlines administrative throughput by automating match progression and participant data handling for 200+ users.
Network Monitoring & Management	PRTG Network Monitor / SolarWinds	Essential for monitoring real-time bit rates and throughput to ensure the network operates within the Shannon Capacity of a noisy channel, specifically targeting the reduction of attenuation, distortion, and crosstalk.
Server Operating Systems	Windows Server 2022 (Standard Edition)	Provides a stable, enterprise-grade L3 environment required for hosting high-availability Game, Web, and FTP servers with necessary redundancy.

3. Specialized Hardware for Services

Performance Rack Servers In accordance with the computer system summarization standards established in Lab Activity 1 (Table 7.2), we specify the deployment of high-density rack servers. These units must feature **Multicore Processors** and a **minimum of 16GB–32GB of RAM** to successfully host the intensive processing requirements of dedicated Game Servers, FTP services, and Live Streaming encoding.

Executive Summary: High-performance multicore servers are the backbone of our "Zero-Downtime" strategy, ensuring that the Data Rate Limit is never breached during peak tournament concurrency.

Monitoring Station This station comprises dedicated administrative workstations equipped with high-resolution multi-monitor arrays. The hardware must be capable of processing complex **Frequency Domain** visualizations provided by PRTG/SolarWinds, allowing engineers to identify signal spikes, interference, or bandwidth saturation in real-time.

Executive Summary: The Monitoring Station facilitates proactive Latency Mitigation, allowing the technical team to visualize the frequency spectrum and resolve impairments before they impact the competitive field.

4. Connectivity & Audio-Visual Enhancements

Enhancement Item	Specification & Role
VoIP Coordination Headsets	Low-latency communication hardware for broadcasters and administrators to ensure synchronized event orchestration.
LED Scoreboard Displays	Large-scale high-refresh-rate displays positioned for the Spectator Role (as defined in Section 2.3 of the Project Spec) to present live match telemetry.

Audio-Visual Impact and Complexity Analysis

The integration of specialized VoIP and LED display systems significantly elevates the project's complexity by requiring advanced traffic management. To ensure that high-definition broadcast streams and real-time audio do not collide with critical game data on the primary backbone, we implement **Multiplexing principles (specifically TDM/Time-Division Multiplexing)**. This ensures that multiple signal components are combined over a single link without inter-channel interference, directly fulfilling the "Complexity of the Project" and "Overall Experience" scoring criteria in the Project Rubric.

5. Integrated Costing Summary & Minimal Cost Justification

Category	Estimated Service Total (RM)
Software Licensing (Management & Monitoring)	RM 850.00
Specialized Hardware (Rack Servers & Monitoring Station)	RM 14,500.00
Connectivity & AV Enhancements (VoIP & LED Displays)	RM 4,200.00
Grand Total for Additional Services	RM 19,550.00

Justification of Expenses

1. **Mitigation of Signal Impairments:** These investments are vital to detect and rectify attenuation, noise, and distortion. Without these tools, the network would be susceptible to signal loss, compromising the professional standard required by UTeM FTMK.
2. **Competitive Integrity & Propagation Time:** Maintaining a strict **Propagation Time** (Source 34.0) is essential for fair play. Dedicated hardware ensures that the time required for a bit to travel through the medium is minimized, preventing latency-driven advantages/disadvantages between players.
3. **Data Rate Limit Adherence:** By investing in specific server tiers and multicore architectures, we ensure the network capacity (C) meets the demands of 200+ users. This aggressive approach to infrastructure protects the regional tournament's reputation from the catastrophic failure of "Noisy Channels" under heavy load.

6. Implementation Readiness

The project team is fully prepared to implement these services, leveraging the technical competencies acquired during the BAXZ 1313 lab sessions. We will deploy an **appropriate Private IP Addressing Scheme using VLSM (Variable Length Subnet Masking)** within a Class C range to securely segment gamers, spectators, and administrators. Our proven skills in Ethernet cable construction, resource sharing, and TCP/IP troubleshooting ensure that this high-performance network will be delivered with professional-grade stability and security.